



OLIST DATASET ANALYSIS

TECHNICAL EXAM AND CASE STUDY

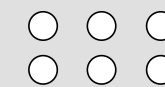
P R E S E N T A T I O N

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Introduction

This report provides a comprehensive analysis of the Brazilian E-Commerce Public Dataset provided by Olist, the largest department store in Brazilian marketplaces

Objectives

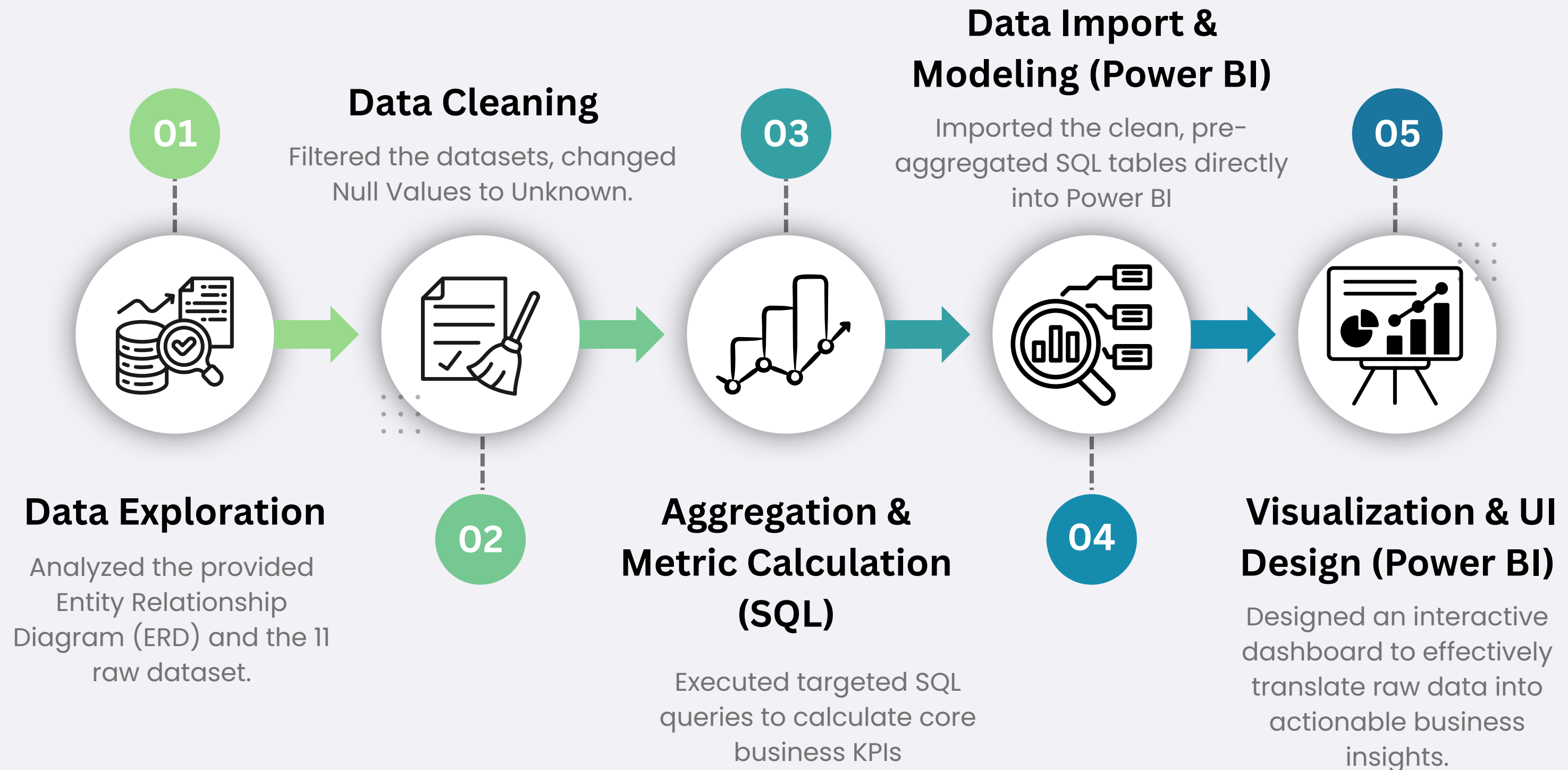
Sales Performance Evaluating revenue trends and category profitability.

Logistical Efficiency Identifying geographical bottlenecks and SLA (Service Level Agreement) compliance.

Sales Performance Determining the critical factors that influence customer satisfaction.

Methodology

To ensure data integrity, this is the flow utilized for the assessment



Sales Performance Dashboard



Dashboard

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John Carlo Ablay

Junior Data Analyst



Dashboard

Statistics

Reports

Database

Team

Promotion

My Store

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Sales Performance

Sort by ▼ This Month

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Order Year

☐ 2016

☒ 2017

☐ 2018

Monthly Revenue Trend

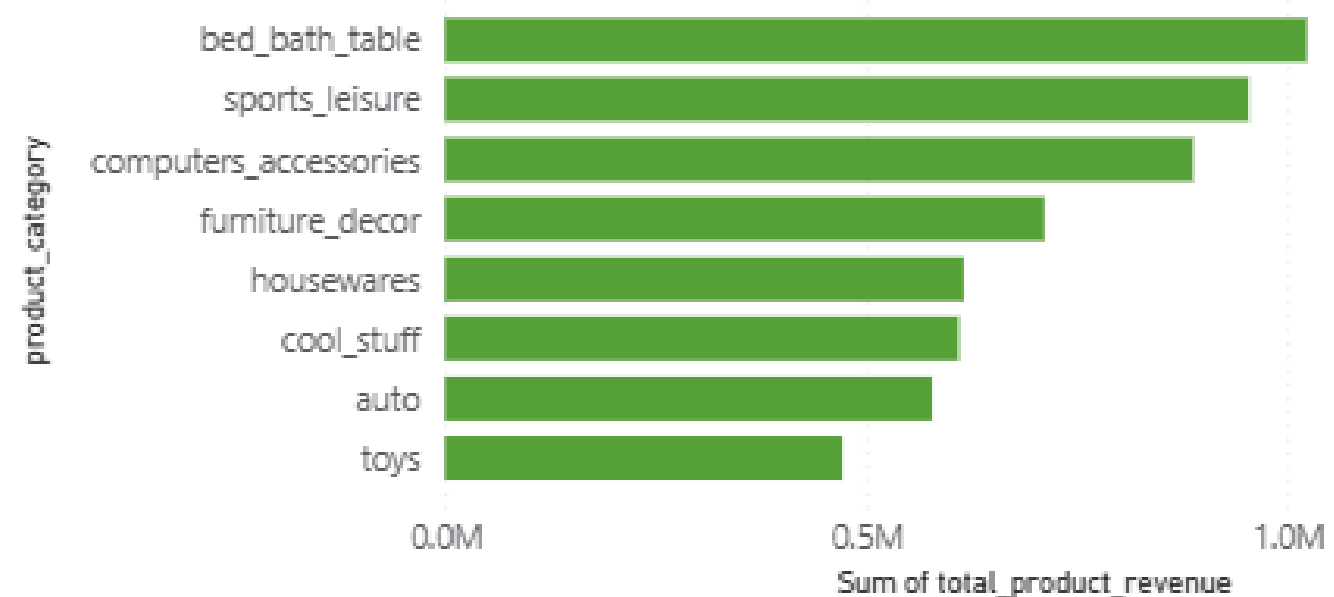


159.86

Average Order Value

43K

Sum of total_orders



Logistical Efficiency Dashboard



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Junior Data Analyst



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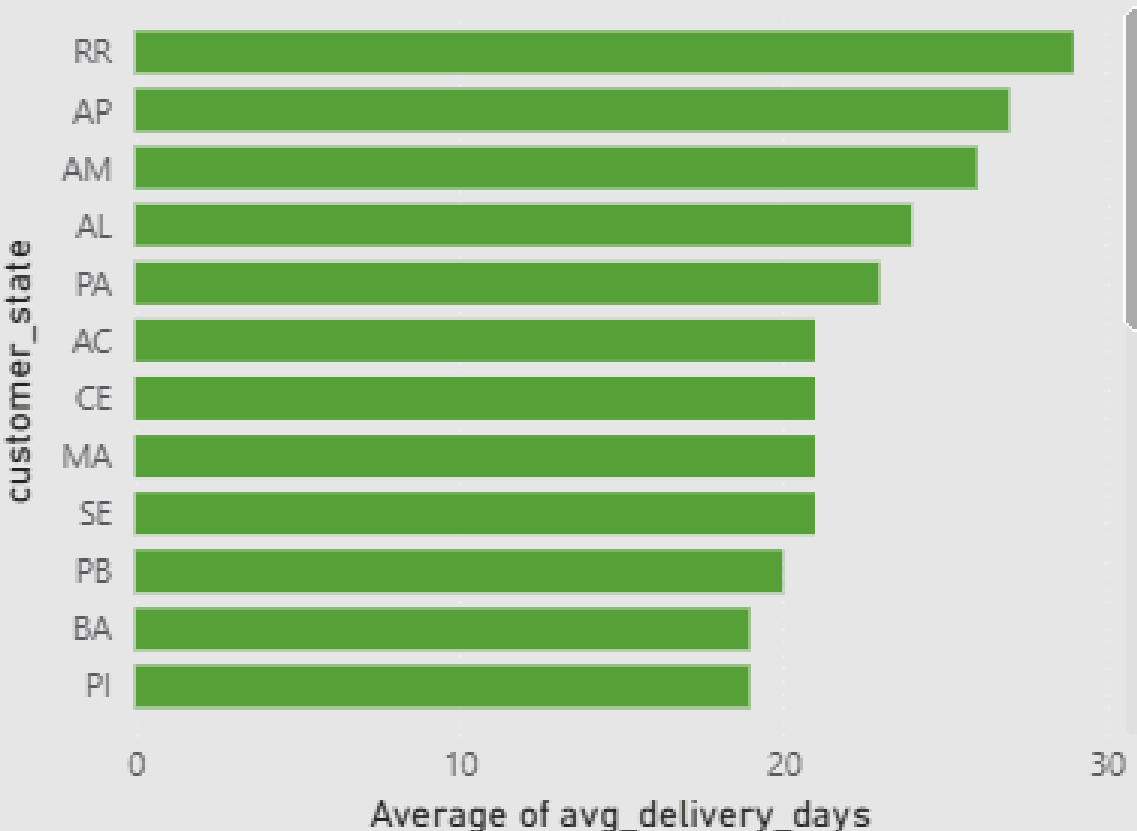
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Logistical Efficiency

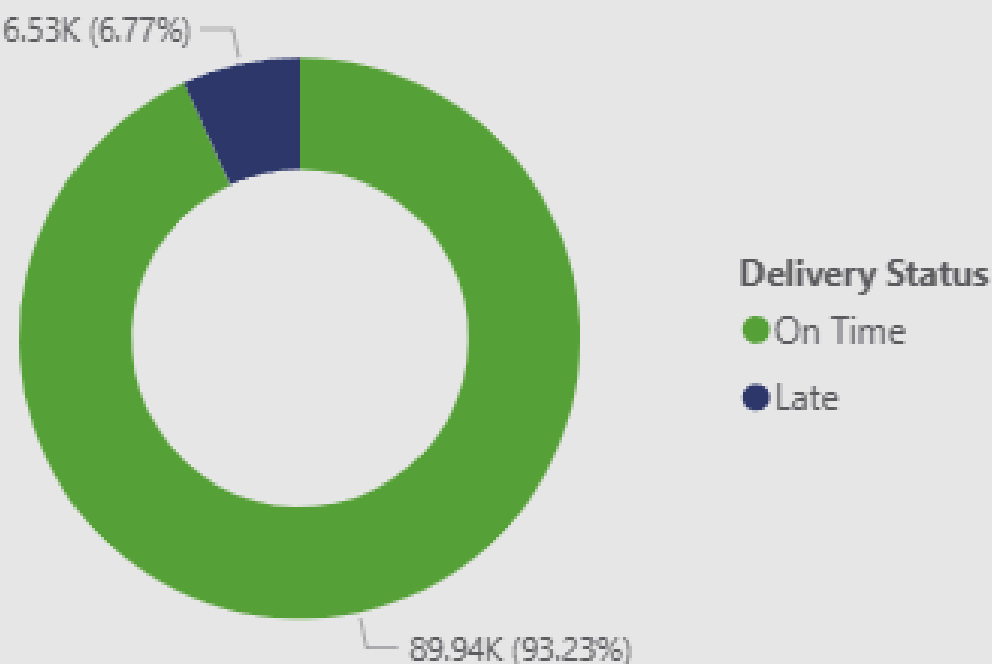
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Sum of Ave Delivery by customer_state



Count of orders by Delivery Status



18.26

Average of avg_delivery_days

96.47K

Count of order_id

Customer Behavior Dashboard



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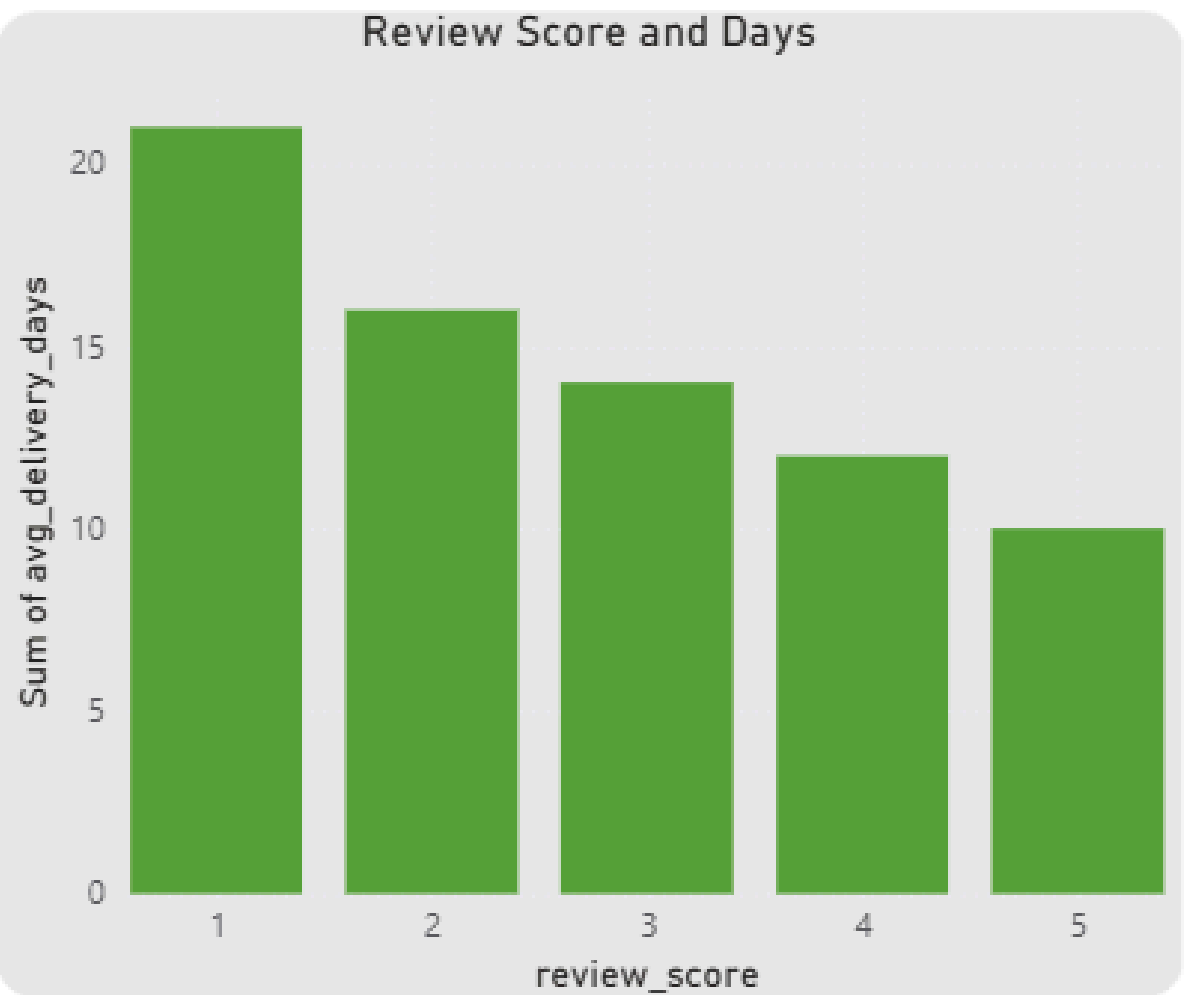
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Customer Behavior

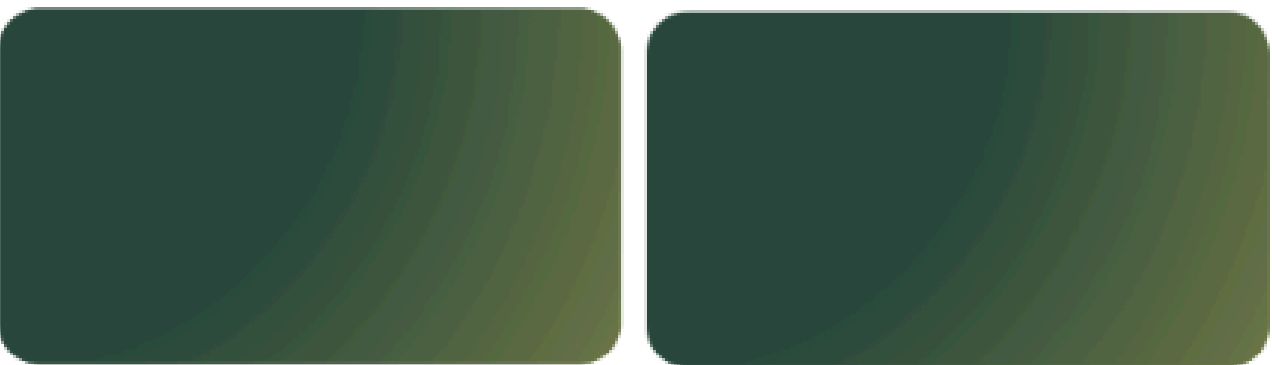
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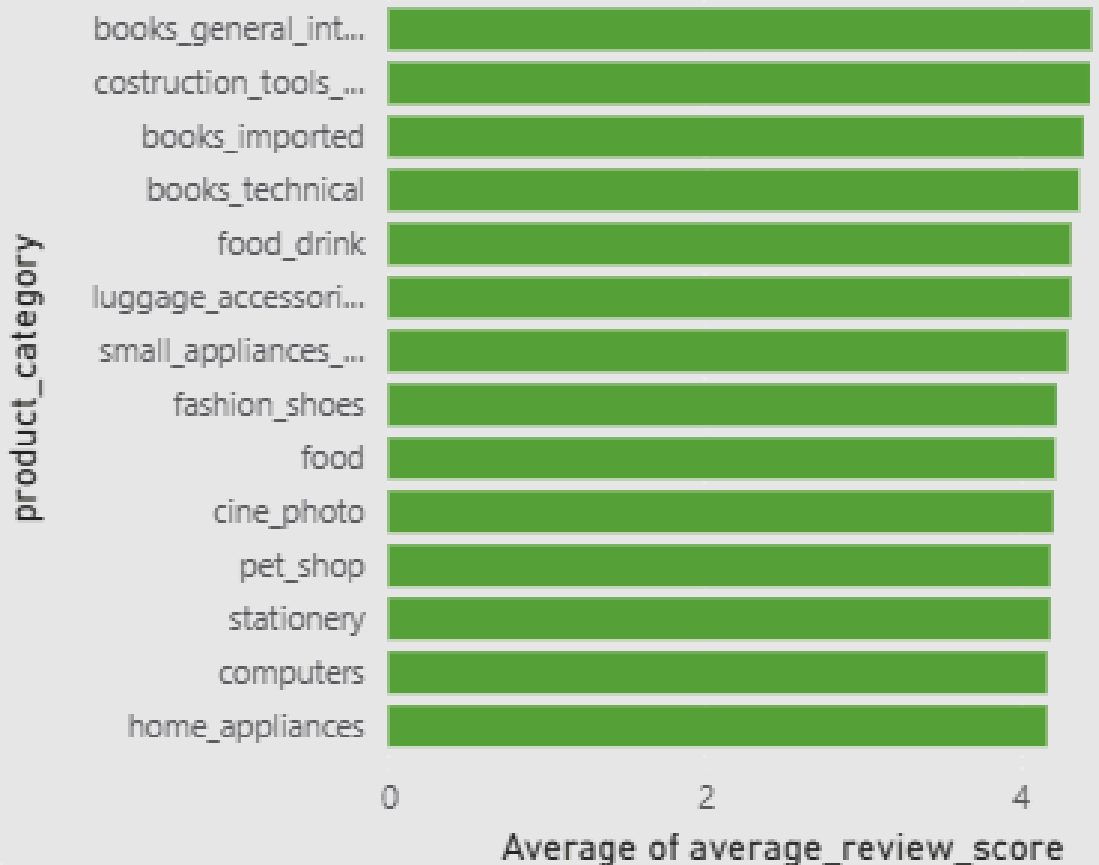


4.07

Average of average_review_score



Average of average_review_score by product_category



Results and Critical Insights

A. Sales Performance



Sustained Revenue Growth

The monthly revenue trend shows a definitive upward trajectory throughout 2017, culminating in a massive spike in **Q4 (November)**.



High-Value Categories

health_beauty, **watches_gifts**, and **bed_bath_table** are the primary revenue drivers.



Healthy Purchasing Habits

Over 43,000 recorded orders, the Average Order Value (AOV) sits at a robust **159.86 BRL**, indicating strong consumer trust in purchasing mid-tier goods.

Results and Critical Insights

B. Logistical Efficiency



Strong Platform Baseline

Olist maintains a highly successful **93.23%** On-Time delivery rate across **96.47K** total shipments.



Geographical Bottlenecks

The data reveals severe regional disparities. **Northern states** such as Roraima (RR), Amapa (AP), and Amazonas (AM) experience the longest average delivery times, heavily skewing the platform's overall delivery metrics.

Results and Critical Insights

C. Customer Behavior



The "Speed-to-Satisfaction" Correlation

5-star reviews average roughly **10 days** for delivery, whereas 1-star reviews are heavily associated with delivery times stretching to 20 days or more.



Product Quality Satisfaction

Overall platform rating is a strong **4.07**. Categories like **books** and **construction_tools** yield the highest customer satisfaction, suggesting high vendor quality in these segments.

Recommendations

Based on the analysis, the following strategic actions are recommended to optimize platform performance



01

Strategic Logistics Restructuring

Olist must establish partnership with localized **Third-Party Logistics (3PL)** providers in the Northern states (RR, AP, AM) to drastically cut the 20+ day lead time



02

Capitalize on Top Categories

Olist should implement targeted **cross-selling campaigns** (e.g., bundling items) to push the 159.86 BRL Average Order Value even higher.



03

Further Analysis

Future analysis should investigate the correlation **between freight value (shipping cost) and cart abandonmentts**

Conclusion

Olist demonstrates excellent market penetration, evidenced by strong revenue growth and a high volume of orders. The platform successfully fulfills over **93%** of its deliveries on time. However, the data clearly dictates that **logistical friction in remote states** is the single largest threat to customer satisfaction. By addressing the fulfillment bottlenecks in the North and leaning into the marketing of high-margin categories like Health & Beauty, Olist can successfully elevate its **4.07** average review score and drive greater customer lifetime value.